

Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

Unit 6 assessment

Unit summary

The browser asks for, receives and draws web pages; the search engine is itself a site keeping an index of the web, and its first result mixes relevance with payment - so search deliberately (specific words, minus signs, quotes, site limits) and read five titles before tapping. Communication tools divide by purpose: email for the formal record, chat for speed, video for live distance work, forums for extended discussion, platforms for structured courses. Safe browsing rests on reading the domain (the part before the first single slash, the one thing an attacker cannot fake cheaply), knowing the padlock means encryption not honesty, judging demands, and reaching account pages by a known address. Artificial intelligence learns patterns from vast examples and predicts - fluent, sometimes fluently wrong, so AI drafts and the teacher decides under UNESCO's enhance-not-replace rule. The Internet of Things embeds sensing in ordinary objects, automating records at the price of automatic collection of often-sensitive data. Blockchain is a chained, sealed, distributed ledger whose old entries cannot be silently changed - earning its cost only where distrusting parties must share a resistant record. Virtual reality replaces the world; augmented reality adds to it; both show engagement gains and both must answer the closing question: what does this do for this lesson, at this cost, with these cautions?

Summative assessment

Multiple-choice questions (MCQs)

1. The difference between a browser and a search engine is: (*Understand*) a) there is none; they are two names for one thing b) the browser asks for and

draws pages; the search engine is a site that finds pages c) the search engine draws pages; the browser finds them d) browsers work only on phones

2. A better search for Class 6 photosynthesis worksheets from Pakistani institutions is: *(Apply)* a) plants b) photosynthesis worksheet c) photosynthesis Class 6 worksheet site:edu.pk d) science
3. Which tool fits a formal record a school might need to prove later? *(Apply)* a) instant chat b) email c) a phone call d) a status update
4. In the address "fee-portal-pk.com/verify", the part that identifies the site's owner is: *(Remember)* a) "verify" b) "fee-portal-pk.com" c) the word "fee" d) the padlock icon
5. The padlock next to an address tells you: *(Understand)* a) the site is honest b) the site is government-run c) the connection is encrypted d) the page is safe for children
6. AI systems can be "confidently, fluently wrong" because they: *(Understand)* a) are broken most of the time b) predict patterns from examples rather than understand as people do c) use old computers d) are programmed to lie
7. A biometric attendance device is best described as: *(Apply)* a) an AI drafting tool b) an Internet of Things object that senses and reports automatically c) a blockchain d) a search engine
8. The key property of a blockchain is: *(Remember)* a) it is always faster than a database b) old entries cannot be silently changed without breaking every later seal c) it needs no electricity d) it cannot hold text
9. The difference between VR and AR is: *(Understand)* a) VR adds to the real world; AR replaces it b) VR replaces the world; AR adds digital objects to the real one c) VR is for games only; AR is for schools only d) there is no difference
10. UNESCO's framing rule for AI in education is: *(Remember)* a) AI should replace teachers to save money b) AI should enhance human capacity

rather than replace teachers, with human accountability c) AI should decide pupils' grades d) AI should not be mentioned in schools

Restricted-response questions (RRQs)

1. Explain why the first search result is "a bid as often as a judgement".
(*Understand*)
2. Plan a search for past papers for Class 8 science from the Sindh board: write the search and one rule you applied. (*Apply*)
3. Match each need to a tool with one clause of reason: a formal report to parents; a quick question to a colleague; a weekly class discussion. (*Apply*)
4. The address "rewards.easypaisa-verify-pk.com/claim" arrives by SMS. Analyse the domain and state the safe action. (*Analyze*)
5. Explain what the padlock does and does not tell you, in two sentences.
(*Understand*)
6. Explain what makes AI drafting useful AND what rule its fluent errors demand. (*Understand*)
7. A school installs a fingerprint attendance system. Name the data it collects and the four questions the school must answer. (*Apply*)
8. A vendor proposes blockchain for the school's internal attendance register. Analyse whether it needs one, using the trust question. (*Analyze*)
9. Distinguish VR from AR with one classroom example of each. (*Understand*)
10. A review finds AR and VR show "engagement gains" but "maturing evidence of learning gain". Analyse what a school should conclude from that pair of findings. (*Analyze*)

Extended-response questions (ERQs)

1. Prepare the one-page "search and communicate" guide for a colleague: five search rules with a worked example, and a tool-matching table for five school communication needs. (*Analyze*)
2. Design the front of a "read the address" card for the staff room: finding the domain with a worked example, the four checks in order with their limits, and the two sentences that defeat most fake-portal messages. (*Analyze*)
3. Write a one-page emerging-technology briefing for your head teacher on an AI drafting tool OR a biometric attendance system: what it is, the school

use, the cautions tied to named rules, and a recommendation with conditions. (*Analyze*)

4. Assess the three fixed-budget offers (four-headset VR lab, phone-based AR science pack, blockchain certificate service) and recommend one with conditions. (*Evaluate*)
5. Using the whole course, evaluate the statement: "A teacher who can name a technology can be sold anything; a teacher who can ask three questions about it cannot." Draw on at least three of the course's units as evidence. (*Evaluate*)

Answers and marking guidance

MCQ answer key

1. b - the browser carries you; the search engine finds.
2. c - specific words plus a site limit beats one word.
3. b - email is formal and keeps records.
4. b - the domain is the part before the first single slash.
5. c - the padlock means encryption; fake sites can have one too.
6. b - prediction from patterns, not human understanding, explains fluent error.
7. b - sensing plus connection in an ordinary object is IoT.
8. b - the chained seals are the property; speed and cost are not it.
9. b - VR replaces; AR adds.
10. b - enhance, not replace, with human accountability.

RRQ model answers and mark schemes

1. Model: search ranking mixes relevance and popularity with paid placement, so the top spot is often bought; reading several titles and addresses beats tapping the first. **Marks (2):** 1 for paid placement; 1 for the read-first habit.
2. Model: "Class 8 science past papers Sindh board site:edu.pk" (or gov.pk); the rule: specific words plus a site limit. **Marks (2):** 1 for a workable search; 1 for the named rule.
3. Model: parents get email (formal, provable); the colleague gets chat (fast, informal); the class gets a discussion forum (extended, every voice). **Marks**

(3): 1 per match with a reason.

4. Model: the domain is "easypaisa-verify-pk.com", not the company's real domain - the right words in the wrong place; the action: do not tap, and reach the real service by typing its known address. **Marks (3):** 1 for the domain; 1 for the mismatch; 1 for the action.
5. Model: the padlock means the connection is encrypted so intermediaries cannot read it; it says nothing about the site's honesty, because fake sites can have padlocks too. **Marks (2):** 1 per half.
6. Model: drafting is useful because it produces fluent material in minutes; the errors demand the rule that AI drafts and the teacher reviews every fact before anything reaches a pupil. **Marks (2):** 1 per half.
7. Model: it collects fingerprints (biometric, sensitive) plus times and presence; the four questions: why collected, where stored, who can see it, how long kept. **Marks (3):** 1 for the data; 2 for the four questions.
8. Model: a single school's internal register has no cross-party trust problem - one office keeps it and the school trusts itself - so a blockchain's cost buys little; district-wide records that must survive any single office might justify one. **Marks (3):** 1 for reading the trust situation; 2 for the analysis.
9. Model: VR replaces the world (headset walk through a virtual Mohenjo-daro); AR adds to it (phone labels on real lab equipment). **Marks (2):** 1 per technology with an example.
10. Model: engagement is real but not the goal; the school should pilot short, curriculum-anchored uses and measure learning, not excitement, before spending - especially where devices are costly. **Marks (3):** 1 for separating the findings; 2 for the implication.

ERQ rubrics

1. **Search and communicate guide (Analyze).** Criteria: *Five distinct workable search rules (0-3); Worked example shows a planned search and judged results (0-4, Analyse); Tool-matching table fits purposes with reasons (0-2); One page (0-1).* Total 10.
2. **Read-the-address card (Analyze).** Criteria: *Finding the domain explained with a worked example (0-3); Four checks in order, each with its limit (0-4, Analyse); The two sentences are the right two (0-2); Fits a card (0-1).* Total 10.

3. **Emerging-technology briefing (Analyze).** Criteria: *Technology explained in plain language (0-2); School use concrete and realistic (0-2); Cautions tied to named rules (0-4, Analyze); Recommendation carries conditions (0-2).* Total 10.

4. **Three offers (Evaluate).** Criteria: *Each offer explained accurately (0-3); Each assessed with the right tool - trust question, lesson-fit question (0-4, Analyze); Recommendation with conditions and budget rationale (0-2); One page (0-1).* Total 10.

5. **Integrative evaluation (Evaluate).** Criteria: *Draws on at least three course units (0-4); Each drawn part used accurately (0-3); The three-questions judgement defended as a synthesis (0-2); Clarity (0-1).* Total 10.

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